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Case 20-2026: A 38-Year-Old Man with Abdominal Pain

Aima A. Ahonkhai, M.D., M.P.H.,^{1,2} Howard M. Heller, M.D., M.P.H.,^{1,2}
and Sarah M. Schrader, M.D., Ph.D.³⁻⁵

PRESENTATION OF CASE

Dr. Howard M. Heller: A 38-year-old man was evaluated at an outpatient clinic of this hospital because of abdominal pain.

The patient had been in his usual state of good health until 6 weeks before the current presentation, when watery diarrhea (up to 20 episodes per day), accompanied by fever, chills, nausea, and vomiting, developed during a 3-week trip to rural Madagascar for work. Episodes of diarrhea started approximately 1 week after he arrived in Madagascar and lasted for 3 days, during which time he had limited food intake. The symptoms then gradually resolved, and he reported feeling well for 3 days afterward; however, the diarrhea then recurred. He received a single oral dose of levofloxacin, and the diarrhea resolved after 2 days. He returned to Massachusetts and then took a brief trip to Texas.

Two weeks after returning from Madagascar, the patient began to have episodes of postprandial nausea, abdominal cramping and pain, bloating, and gas. The episodes resolved spontaneously within approximately 5 to 7 hours. No diarrhea was associated with these episodes. His symptoms did not worsen after ingestion of dairy, grains, fruits, or vegetables, but he recalled that an episode of cramping pain and bloating occurred after he had eaten a hamburger.

Approximately 2 weeks after the episodes began, the patient presented to the outpatient clinic of this hospital. A review of systems was notable for an unintentional weight loss of 3 kg and fatigue that had been present for several weeks. He had no fever, chills, sweats, or respiratory, genitourinary, or neurologic symptoms. He had no known sick contacts. While traveling, he had consumed cooked food (including greens and meats) and only bottled water. During his recent trip to Madagascar, he had worked in forested and aquatic environments, which included wading in rivers. He had also walked barefoot around the field camp. He reported that he had been bitten by mosquitoes but had no other known insect or direct animal exposures; however, his colleagues had handled insects, birds, reptiles, and bats. The patient noted that he had had previous tick exposures, including one tick that had attached to his body in eastern Massachusetts 5 months before the trip to Madagascar, as well as possible tick exposures in South America 1 month before the trip.

Author affiliations are listed at the end of the article.

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Table 1. Laboratory Data.

Variable	Reference Range, Adults*	On Current Presentation
Hemoglobin (g/dl)	13.5–17.5	14.9
Hematocrit (%)	41.0–53.0	43.3
White-cell count (per μ l)	4500–11,000	7690
Differential count (per μ l)		
Neutrophils	1800–7700	3100
Lymphocytes	1000–4800	2100
Monocytes	200–1200	660
Eosinophils	0–900	1700
Basophils	0–300	110
Platelet count (per μ l)	150,000–400,000	240,000

* Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

The patient's medical history included two episodes of malaria, for which he had been treated with mefloquine and atovaquone–proguanil, respectively. Six years before the current admission, a screening test for human immunodeficiency virus (HIV) types 1 and 2 was negative. Medications included a multivitamin and melatonin as needed for sleep; he had taken atovaquone–proguanil for malaria prophylaxis during his recent trip to Madagascar. He had no known adverse reactions to medications. The patient had received vaccines against coronavirus disease 2019, hepatitis A and B, Japanese encephalitis, measles, mumps, polio, rabies, rubella, tetanus, typhoid fever, varicella, and yellow fever, as well as the bacillus Calmette–Guérin vaccine. The patient resided in Massachusetts and was employed as an ornithologist, which often involved fieldwork in southern and coastal Africa, East Asia, South America, and the South Pacific. The family history was unremarkable. He did not use alcohol, tobacco, or illicit substances.

On examination, the temporal temperature was 36.6°C, the heart rate 68 beats per minute, the blood pressure 121/72 mm Hg, and the oxygen saturation 95% while the patient was breathing ambient air. The weight was 86.0 kg. He appeared well, and no lymphadenopathy was present. The abdomen was soft and nontender,

with normal bowel sounds and without organomegaly. He had no rashes.

The blood levels of alanine aminotransferase, aspartate aminotransferase, alkaline phosphatase, bilirubin, albumin, globulin, and total protein were normal. The white-cell count was 7690 per microliter (reference range, 4500 to 11,000), and the eosinophil count was 1700 per microliter (reference range, 4500 to 11,000). Other laboratory test results are shown in Table 1. Specimens of blood and stool were obtained for further testing.

Over the course of the following week, the patient had episodic abdominal pain and increased frequency of bowel movements. During this time, an examination of a stool specimen revealed no ova or parasites, including cyclospora and cystoisospora. In addition, tests for antibodies to schistosoma and strongyloides and stool antigen tests for giardia and rotavirus were negative.

Diagnostic test results were received.

DIFFERENTIAL DIAGNOSIS

Dr. Aima A. Ahonkhai: A limited acute diarrheal syndrome developed in this 38-year-old ornithologist during a recent work trip to Madagascar, and the patient now presents with recurrent postprandial abdominal pain and peripheral eosinophilia. He has a background of residence in Massachusetts and frequent global travel, including trips to Africa, Asia, and the South Pacific. The patient is presumed to be immunocompetent; he is not known to have HIV infection and does not take any immunosuppressive medications. I will approach this case by considering infectious causes of the patient's presentation, his immune status, the timeline of symptoms relative to his travel history, and the relevant epidemiologic factors and exposures before, during, and after travel to Madagascar.

INITIAL DIARRHEAL ILLNESS

The patient's first gastrointestinal symptoms began approximately 1 week after his arrival in Madagascar and included watery diarrhea, nausea, and vomiting, accompanied by fever and chills. Did this patient have traveler's diarrhea? Diarrhea is the most common illness in travelers visiting resource-limited countries, and it occurs

in 30 to 70% of persons after 2 weeks of travel.¹ The incidence varies depending on the level of sanitation in the community, and traveler's diarrhea may still occur despite drinking bottled water and eating cooked foods, as this patient did. Bacterial pathogens are the most commonly implicated cause and account for 80 to 90% of cases. Enterotoxigenic *Escherichia coli* is the most likely cause in this patient with watery diarrhea. Viruses account for 5 to 15% of cases, and protozoa for approximately 10%.¹

Most persons with traveler's diarrhea have self-limited symptoms. This patient's symptoms resolved for a period of several weeks after a single dose of levofloxacin; however, he now presents with recurrent gastrointestinal symptoms, along with postprandial abdominal pain and bloating. Unlike acute infectious diarrhea, which is often caused by bacterial pathogens, chronic diarrhea with upper gastrointestinal symptoms is most likely to be caused by protozoal pathogens such as *Giardia lamblia*, *Cyclospora cayetanensis*, *Cystoisospora belli*, and *Dientamoeba fragilis*. *Giardia* antigen testing, which has high sensitivity and specificity, was negative in this patient.² Although the sensitivity and specificity of stool microscopy is variable and highly dependent on the microscopist's skill, it was also negative, a finding that makes other protozoal pathogens unlikely causes of this patient's symptoms. In addition, neither *G. lamblia* nor *C. cayetanensis* infection would explain the eosinophilia in this patient.

Persistent symptoms and negative testing for protozoal pathogens suggest the possibility of a noninfectious cause of the patient's gastrointestinal symptoms. Among persons with traveler's diarrhea, 6 to 18% have chronic gastrointestinal symptoms, a feature that may reflect unmasking symptoms of underlying gastrointestinal diseases (e.g., celiac disease or inflammatory bowel disease), postinfectious malabsorption due to lactase deficiency or bacterial overgrowth, or postinfectious irritable bowel syndrome.^{1,2} However, common infectious causes should be ruled out before considering these diagnoses, none of which would explain this patient's eosinophilia.

HELMINTHIC INFECTIONS

Eosinophilia was a notable finding in this patient's laboratory evaluation. It is observed in 8 to 10% of travelers returning from resource-

limited countries, and helminths are implicated in 14 to 64% of those cases.³ Several helminthic infections that are associated with gastrointestinal symptoms and are common across Africa, including Madagascar, may provide a unifying diagnosis in this patient.

The most common helminthic infection worldwide is ascariasis, which is caused by *Ascaris lumbricoides*.⁴ It is spread through the fecal-oral route; larvae migrate through the liver and bloodstream into the lungs before being swallowed into the gastrointestinal tract, where they mature into adult worms. The life cycle of ascariis, characterized by the maturation of adult worms in the gastrointestinal tract, may explain why gastrointestinal symptoms develop 6 to 8 weeks after initial ingestion of infective eggs. However, gastrointestinal symptoms are more typical among persons living in regions where the disease is endemic than travelers, such as this patient, and among those with a heavy burden of gastrointestinal worms.

Fascioliasis is an infection caused by *Fasciola hepatica* or *F. gigantica*, which are liver flukes that undergo lengthy, complex life cycles involving snails as intermediate hosts. Human infection ultimately occurs after ingestion of vegetation contaminated with metacercariae. Once ingested, the fluke burrows through the small intestine into the peritoneal cavity and then into the bile ducts. Infection typically occurs after ingestion of uncooked contaminated vegetables, and this patient reported eating only cooked vegetables. In addition, he did not have pain in the right upper quadrant or abnormal alanine aminotransferase or aspartate aminotransferase levels.

Hookworm infection is caused by *Necator americanus* or, less commonly, *Ancylostoma duodenale*. It is acquired when the larvae burrow through skin that has been in direct contact with contaminated soil. This patient may have become infected while he was walking barefoot around the camp in Madagascar; however, given his extensive previous global travel, acquisition from another geographic setting is also possible. Clinically significant hookworm infection is often associated with epigastric abdominal pain, iron-deficiency anemia, and manifestations of chronic intestinal blood loss or malnutrition, particularly in patients with heavy parasite burdens. Although hookworm infection could account for

the eosinophilia and gastrointestinal symptoms in this patient, I would place it lower than other helminthic infections on the differential diagnosis, given the absence of anemia and other evidence of chronic blood loss despite several weeks of symptoms and weight loss.

Schistosomiasis is an important consideration in this patient who spent time wading in freshwater rivers.³ Schistosomiasis is endemic across Africa and hyperendemic in Madagascar, which has the fifth-highest schistosomiasis rate globally and a seroprevalence of 52% (including *Schistosoma mansoni*, *S. haematobium*, and *S. japonicum* infections).⁵ Larvae living in fresh water penetrate the skin, and then worms mate in the mesenteric circulation and produce eggs that can be shed into the gastrointestinal tract, which leads to an inflammatory response and gastrointestinal symptoms. Katayama fever, an acute syndrome characterized by fever, cough, rash, diarrhea, and headache, develops in approximately half the persons with schistosomiasis. Katayama fever is an immune-mediated response to tissue migration and the egg-laying phases of infection and thus occurs 2 to 8 weeks after exposure in fresh water. Although it is tempting to speculate that this patient's episode of watery diarrhea that occurred during travel to Madagascar was a symptom of Katayama fever, onset of this syndrome after approximately 1 week in Madagascar would be rather early.

Strongyloidiasis, an infection caused by *Strongyloides stercoralis*, is spread through direct contact of the skin with contaminated soil. Larva currens syndrome, characterized by pruritus or erythema at the site of inoculation, occurs in some persons with strongyloidiasis, but such symptoms were not reported in this patient. Gastrointestinal symptoms occur after adult worms develop in the gastrointestinal tract, typically 2 weeks after the initial infection. Although strongyloidiasis remains an important consideration, given the patient's barefoot soil exposure, I would consider it less likely than schistosomiasis in this case. Strongyloidiasis is endemic in Madagascar, but schistosomiasis is hyperendemic and poses a substantially greater epidemiologic risk, particularly among persons with freshwater exposures. Given this patient's history of freshwater contact, schistosomiasis would rank higher on the differential diagnosis.

Diagnosing these helminthic infections can be challenging because antibody or antigen tests and microscopic examination of stool specimens, when performed early in the infection, are often negative. Tests for helminth-specific antibodies are often positive before eggs are detectable in the stool.⁶ In this patient, microscopic examination of the stool specimen and serologic testing for schistosomiasis and strongyloidiasis were negative. Repeat testing could be considered after 6 weeks, at which point a serologic test would be expected to be positive. However, given the initial negative test results, I will consider other possible causes of this patient's presentation.

OTHER CAUSES OF EOSINOPHILIA

Several viral, bacterial, mycobacterial, and fungal infections can cause eosinophilia. I would perform testing for HIV, but other infectious causes of eosinophilia would be unlikely to explain the patient's postprandial abdominal pain. Similarly, noninfectious causes of eosinophilia, such as allergic diseases, cancer, or drug reactions, would be unlikely to account for his gastrointestinal symptoms.

PRETRAVEL EXPOSURES

When evaluating a returning traveler, it is important to consider epidemiologic factors and exposures unrelated to travel. This patient's pretravel exposures included living in Massachusetts, where he had had one reported tick bite 5 months before he traveled to Madagascar. This tick exposure in the northeastern United States puts him at risk for alpha-gal syndrome, a disease that could explain postprandial abdominal pain, bloating, and eosinophilia.

ALPHA-GAL SYNDROME

Alpha-gal syndrome is an IgE-mediated delayed hypersensitivity to galactose- α -1,3-galactose (alpha-gal), a carbohydrate found in mammalian meat but absent from human tissue. The condition is initiated by exposure to alpha-gal through tick bites, particularly those from the lone star tick (*Amblyomma americanum*), which can lead to sensitization. Patients with alpha-gal syndrome can present with a broad spectrum of clinical features that ranges from isolated gastrointestinal

symptoms to cutaneous manifestations (e.g., urticaria, pruritus, and angioedema) and severe systemic reactions, including anaphylaxis and, in rare cases, death from cardiovascular collapse. Delayed onset of symptoms (3 to 8 hours) after consumption of mammalian meat products is a notable distinction from other food-borne allergens. Alpha-gal syndrome can develop several weeks after the initial tick exposure, and *A. americanum* has become more common in the northeastern United States during the past decade — a factor that increased recognition of the alpha-gal syndrome, expansion of diagnostic testing, and the geographic expansion of tick vectors. The Centers for Disease Control and Prevention has estimated that alpha-gal syndrome might have developed in 96,000 to 450,000 persons in the United States since 2010.⁷

Given a compatible clinical syndrome, a possible exacerbation with meat consumption (hamburger), and a known appropriate epidemiologic exposure to a tick bite before travel to Madagascar, alpha-gal syndrome is the leading diagnosis in this patient. Of note, peripheral eosinophilia is not a classic feature of alpha-gal syndrome and suggests the possibility of a concurrent eosinophilic or parasitic process, particularly in a patient with extensive global travel and relevant environmental exposures. Nonetheless, a diagnosis of alpha-gal syndrome would be supported by a positive serologic test for alpha-gal IgE and reevaluation 1 month after dietary elimination of alpha-gal-containing products to ensure symptom resolution.⁸

CLINICAL IMPRESSION

Dr. Heller: Applying the principle of parsimony, I considered the probability that the patient's eosinophilia and abdominal pain were related to each other and also related to the recent trip to Madagascar. I also considered the possibility that his occupational exposures and trip to Madagascar were distractors that were unrelated to his illness. Given that he had noted that his symptoms were sometimes worse after ingestion of meat, I suspected alpha-gal syndrome and obtained a specimen of blood for alpha-gal IgE testing.

CLINICAL DIAGNOSIS

Alpha-gal syndrome.

DR. AIMA A. AHONKHAI'S DIAGNOSIS

Alpha-gal syndrome.

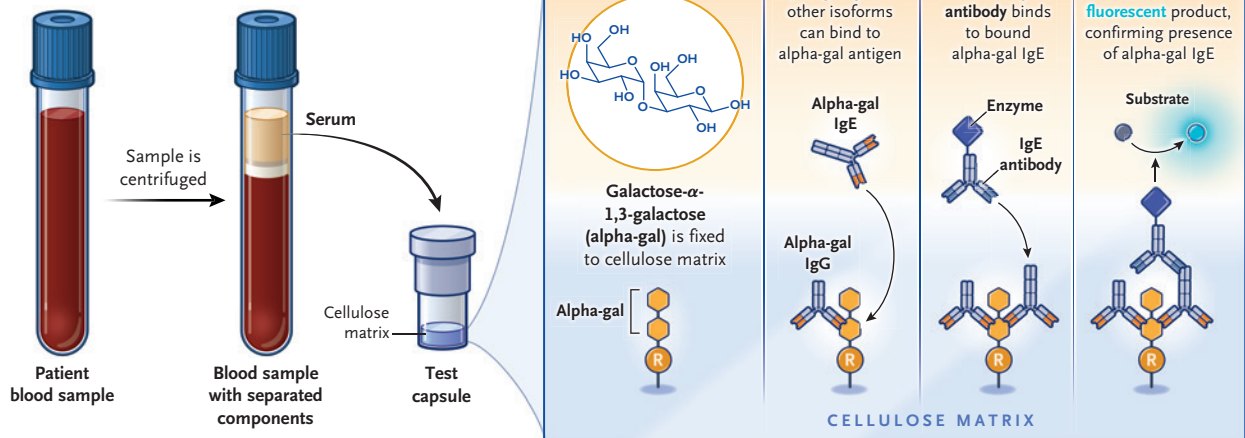
PATHOLOGICAL DISCUSSION

Dr. Sarah M. Schrader: The diagnostic test in this case was an enzyme-linked immunoassay to detect alpha-gal IgE in the patient's serum. The assay yielded a concentration of 13.10 kU per liter (reference value, <0.10), which indicates a high level of alpha-gal IgE reactivity to the test antigen.

Alpha-gal adorns cell-membrane glycoproteins and glycolipids in almost all mammalian species (Fig. 1). Owing to acquisition of a frameshift variant that results in a premature stop codon in one of the enzymes that are required to synthesize alpha-gal, Old World monkeys, apes, and humans do not produce alpha-gal.⁹ Because alpha-gal is not a self-antigen in these species, they can produce antibodies against it and naturally have IgG, IgM, and IgA isoforms owing to exposure to alpha-gal from the gut microbiome, pathogens, and mammalian food products. These antibodies have been associated with protection against viral, bacterial, and parasitic pathogens that express alpha-gal and also constitute a major barrier to xenotransplantation.¹⁰ However, the alpha-gal IgE isoform that was detected in this patient is not naturally occurring.

Alpha-gal IgE was first linked to allergic reactions after the injection of alpha-gal-containing biologic therapies in 2008¹¹ and the consumption of mammalian meat products in 2009.¹² Epidemiologic and experimental evidence has since linked the development of alpha-gal IgE to the bite of ticks of certain species whose saliva contains alpha-gal, including *A. americanum* in North America, *Ixodes holocyclus* in Australia, and *Ixodes ricinus* in Europe. Although the precise mechanisms remain unclear, immunomodulatory factors in tick saliva, together with host factors, most likely lead to sensitization. Once alpha-gal IgE forms, it can bind to IgE

A Assay for Alpha-Gal Syndrome



B Human Acquisition and Development of Alpha-Gal Syndrome

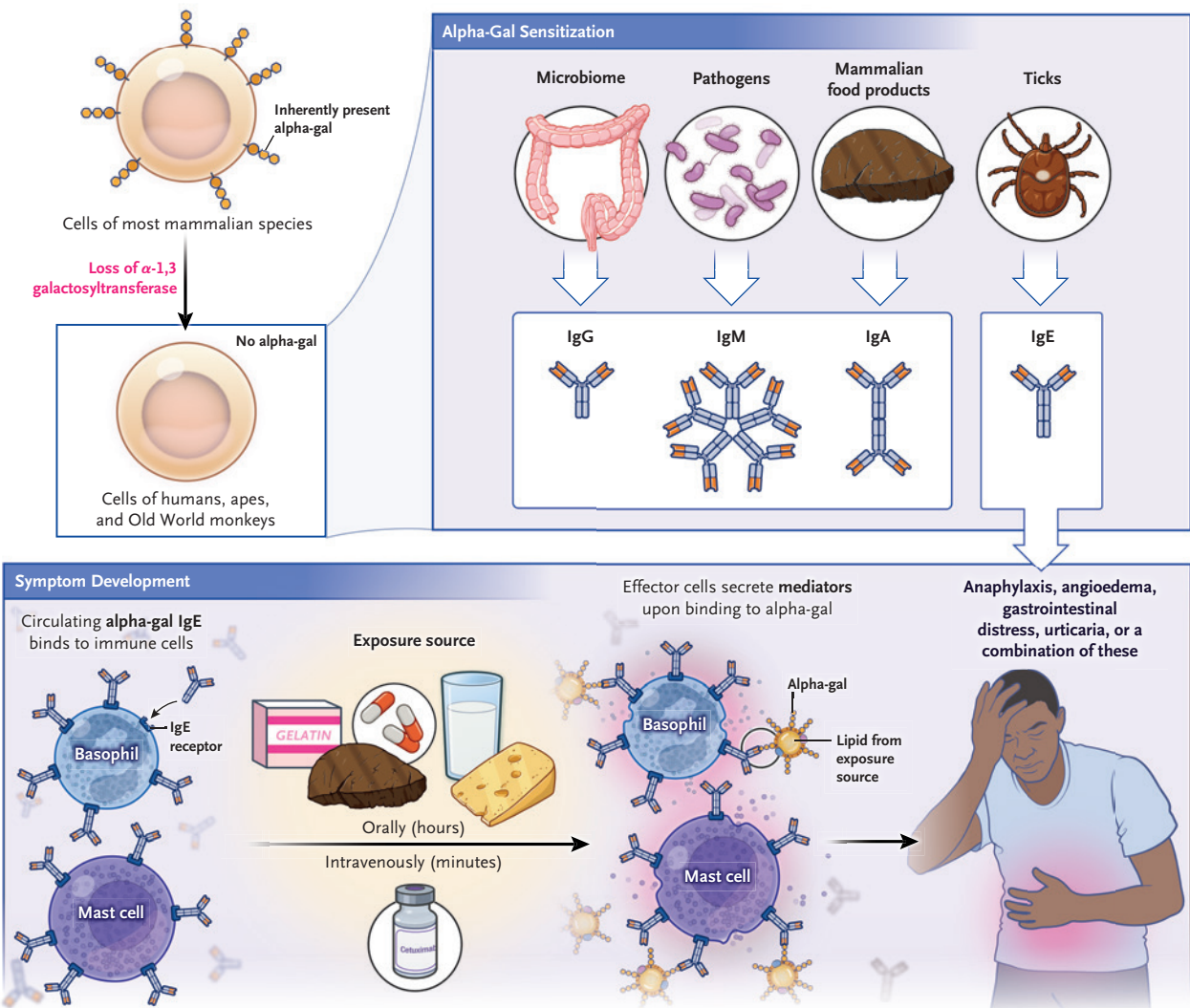


Figure 1 (facing page). Alpha-Gal Syndrome Testing and Pathogenesis.

In the assay for IgE specific to galactose- α -1,3-galactose (alpha-gal) (Panel A), a sample of the patient's blood is centrifuged to separate out the serum component. Serum is then applied to a test capsule containing a cellulose matrix bound to the carbohydrate antigen alpha-gal. Alpha-gal-specific antibodies in the patient's serum bind to the alpha-gal on the matrix while the remainder of the serum contents are washed away. An enzyme-linked reagent antibody specific for human IgE is then added and binds to captured alpha-gal IgE. Finally, the enzyme converts its substrate into a fluorescent product. The resulting fluorescence is measured and converted to an alpha-gal IgE concentration with a calibration curve. The cells of most mammalian species display alpha-gal on their surface (Panel B). Owing to the loss of one of the enzymes required for its synthesis, humans, apes, and Old World monkeys do not produce alpha-gal and are thus able to produce antibodies against it. In these species, alpha-gal IgG, IgM, and IgA are naturally occurring owing to exposure to alpha-gal from the gut microbiome, pathogens, and consumption of mammalian food products. Alpha-gal IgE is produced only after sensitization through the bite from a tick of certain species. The saliva of these ticks contains alpha-gal along with immunomodulatory molecules that combine with host factors to promote alpha-gal IgE production. Once produced, alpha-gal IgE attaches to IgE receptors on the surfaces of basophils and mast cells. When the sensitized individual encounters alpha-gal again, these cells can mediate the development of symptoms, which can range from gastrointestinal distress to anaphylaxis. Symptoms typically develop hours after oral ingestion of alpha-gal and minutes after intravenous exposure. Alpha-gal is found in food products derived from mammalian species (e.g., meats, dairy products, and gelatin). Some biologic therapies (e.g., cetuximab) also contain alpha-gal moieties. R denotes rest of the molecule.

receptors on basophils and mast cells. When a sensitized individual next encounters the alpha-gal moiety (e.g., after ingestion of mammalian food products), these basophils and mast cells can undergo degranulation, which triggers an allergic reaction. This sequence of events is known as alpha-gal syndrome. The patient's symptoms, along with a positive test for alpha-gal IgE, established a diagnosis of alpha-gal syndrome.

PATHOLOGICAL DIAGNOSIS

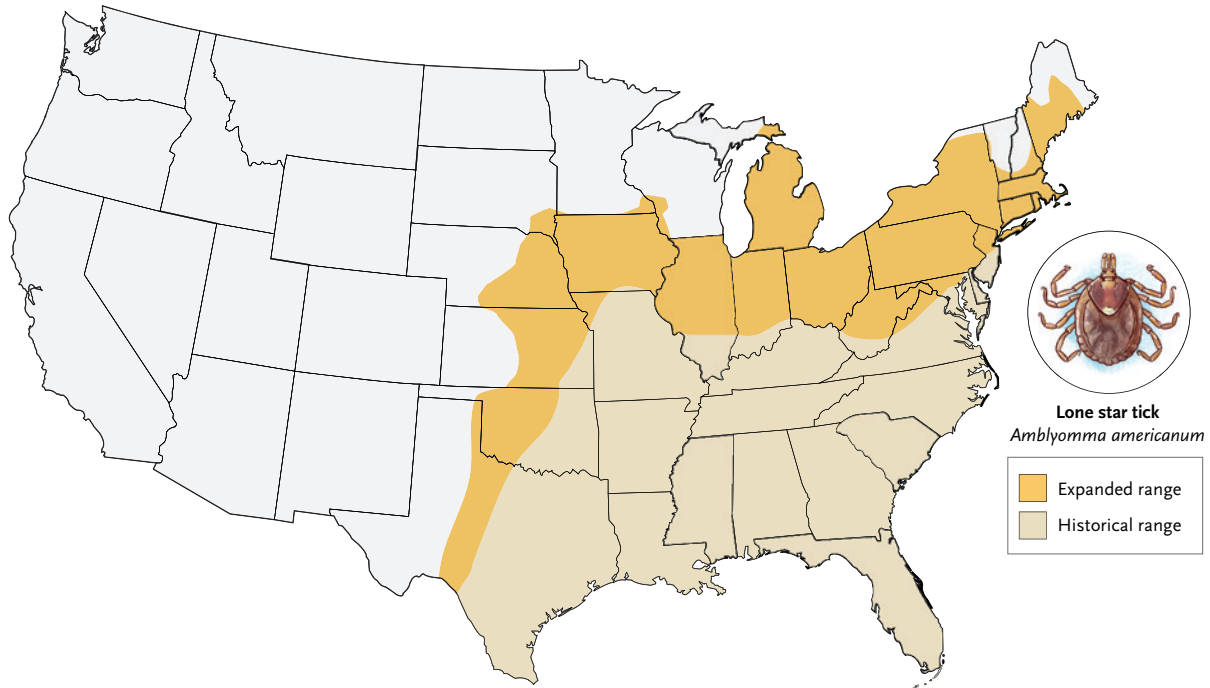
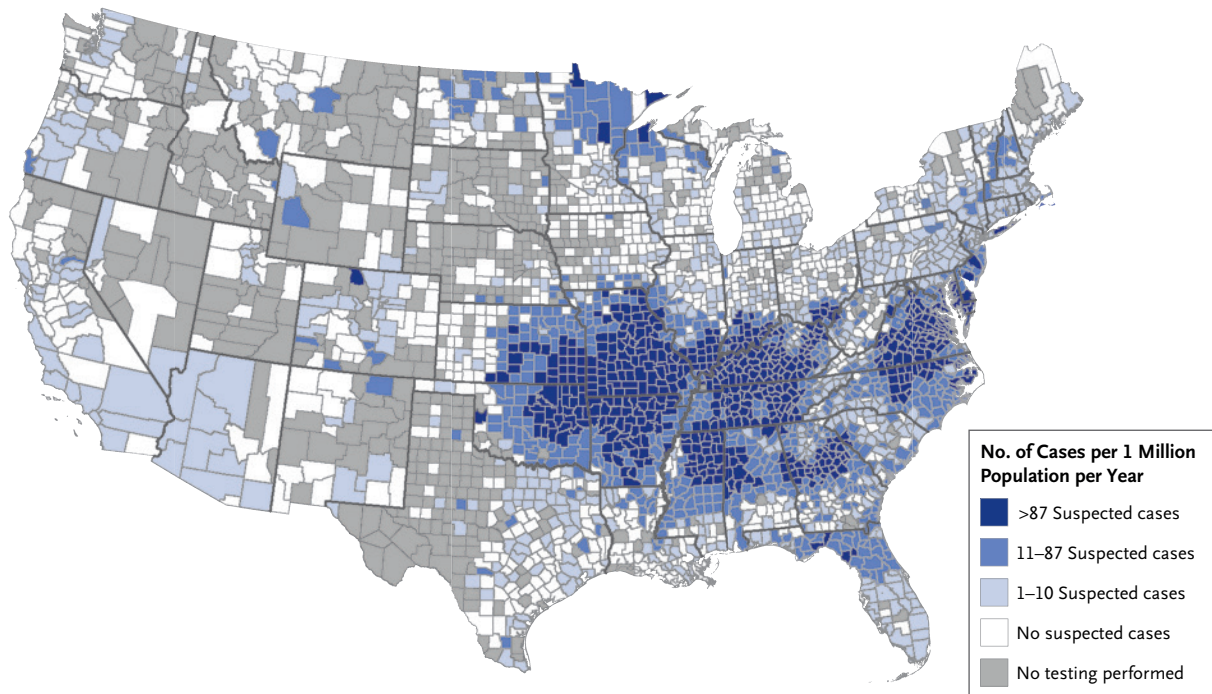
Alpha-gal syndrome.

DISCUSSION OF MANAGEMENT

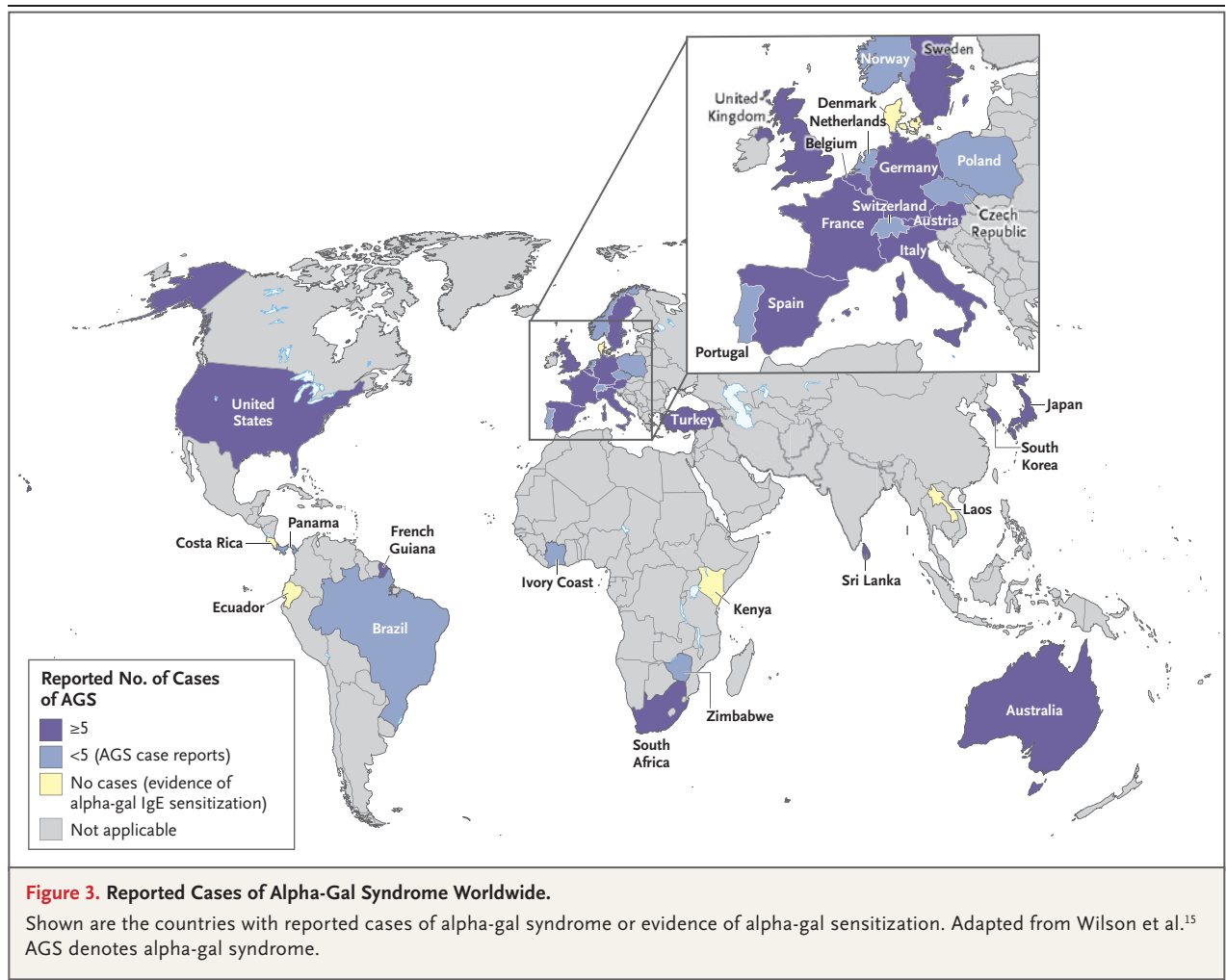
Dr. Heller: The primary consideration in the treatment of alpha-gal syndrome is the avoidance of ingestion of mammalian meat and gelatin products.¹³ This includes products that might contain mammalian alpha-gal carbohydrate in minuscule amounts, including medication capsules, foods or candies that contain gelatin, and animal-derived medications such as heparin, some vaccines, antivenoms, and monoclonal antibody preparations. Treatment with antihistamines and glucocorticoids might help to prevent reactions. For persons who have urticaria, respiratory symptoms, or anaphylaxis, epinephrine injectors should be available. Desensitization to alpha-gal can be attempted under the care of an allergist.

The alpha-gal IgE level can be monitored, and it declines with alpha-gal avoidance. Resensitization can occur with subsequent tick bites or ingestion of alpha-gal. In North America, the tick primarily associated with alpha-gal sensitization is *A. americanum*. The geographic range of this tick has expanded in recent decades, in part owing to climate change (Figs. 2 and 3).^{14,15} Ticks can sometimes establish populations in new geographic areas when they are transported by migrating birds.¹⁶ In addition, warming climates allow for longer seasons during which ticks are active and feeding. The expanding range of *A. americanum* and the increased prevalence of alpha-gal allergy are examples of the relevance of the One Health approach to disease, a holistic, transdisciplinary strategy that recognizes the interconnection among humans, animals, and the environment.

The patient avoided consuming mammalian meat, and his symptoms resolved within 1 month. One year after diagnosis, the patient remains clinically well. The absolute eosinophil count decreased from 1700 to 787 per microliter within 4 weeks after diagnosis and to 316 per microliter after 1 year. The alpha-gal IgE level decreased from 13.10 to 5.88 kU per liter. Eosinophilia has only rarely been reported in association with alpha-gal syndrome,¹⁷ but the resolution of this patient's eosinophilia and symptoms and the decline in the alpha-gal IgE level with avoidance of meat, in the absence of other interventions, suggests that the eosinophilia was related to the alpha-gal allergy.

A Geographic Distribution of the Lone Star Tick, 1945 to 2015**B Suspected Cases of Alpha-Gal Syndrome, 2017 to 2022****Figure 2. Geographic Distribution of the Lone Star Tick and Suspected Cases of Alpha-Gal Syndrome.**

Panel A, adapted from Monzón et al.,¹⁴ shows the geographic range of the lone star tick (*Amblyomma americanum*) in the continental United States over time. The expanded range includes data from 2013, 2014, and 2015, and the historical range includes data from 1945. Panel B shows the geographic distribution of suspected cases of alpha-gal syndrome (2017 to 2022).⁷



PATIENT PERSPECTIVE

The Patient: I do remember being bitten by a tick in Massachusetts sometime in late May. It was a small tick, possibly a lone star tick, that bit me, so it is possible the tick exposure occurred as many as 5 months before I started to develop symptoms, if that was in fact the tick that caused this. Also noteworthy is that I ate red meat in Texas, probably 3 weeks before the diagnosis of alpha-gal syndrome. In the past, I had eaten red meat with no issues, and then I very suddenly started to get abdominal symptoms after eating red meat. In my discussions with Dr. Heller, what tipped us off to this was the clear association between the symptoms and eating pork or beef. I would eat something, and there would be a delay of maybe 1 to 3 hours, then I would have really severe abdominal pain without

diarrhea or rash. The pain would subside maybe 5 to 7 hours later, and I would generally feel very exhausted, sort of like having the flu but without the other flu symptoms. After eating dairy, I did not have abdominal pain but just a general feeling of exhaustion.

Since the diagnosis, I have cut out all red meat, dairy, gelatin — any sort of exposure I can find. I have not had feelings of exhaustion or recurrence of the abdominal pain. I'm grateful to have a diagnosis, and hopefully, this is a condition that resolves.

FINAL DIAGNOSIS

Alpha-gal syndrome.

This case was presented at the Medicine Case Conference.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

¹Department of Medicine, Massachusetts General Hospital, Boston; ²Department of Medicine, Harvard Medical School, Boston;

³Department of Pathology, Brigham and Women's Hospital, Boston;

⁴Department of Pathology, Massachusetts General Hospital, Boston; ⁵Department of Pathology, Harvard Medical School, Boston.

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